

CTMC Models for Customer Journey Prediction

UCLA Stats 148 Capstone — Technical Reference

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Contents

1	Problem Setup	2
2	Model 1: Global CTMC	2
2.1	Overview	2
2.2	Fitting	2
2.3	Prediction	3
3	Model 2: Laplace-Smoothed Global CTMC	3
3.1	Overview	3
3.2	Fitting	3
3.3	Prediction	3
4	Model 3: Clustered CTMC	3
4.1	Overview	3
4.2	Clustering	4
4.3	Fitting	4
4.4	Prediction	4
5	Model 4: Calibrated CTMC	4
5.1	Overview	4
5.2	Isotonic Calibration	4
5.3	Prevalence Shift (Log-Odds Adjustment)	5
6	Model Hierarchy and Composition	5
7	Inference Summary	5

1 Problem Setup

Each customer journey is a sequence of discrete actions $a_1, a_2, \dots, a_n$ recorded at continuous timestamps. The goal is to predict, from a partially observed prefix, whether the journey will reach the success state (action `ed_id` = 28, *order_shipped*) within a 60-day horizon.

We model journeys as realizations of a *Continuous-Time Markov Chain* (CTMC) on the action-type state space $\mathcal{S} = \{s_1, \dots, s_K\}$. The key quantity is the **generator matrix** $Q \in \mathbb{R}^{K \times K}$ with entries

$$q_{ij} = \begin{cases} \lambda_{ij} \geq 0 & i \neq j \\ -\sum_{j \neq i} \lambda_{ij} & i = j \end{cases}$$

where λ_{ij} is the instantaneous rate of transitioning from state i to state j .

Inference target

Given current state s at snapshot time, we want

$$P(T_{\text{success}} \leq \tau \mid X(0) = s) = \left[e^{Q^* \tau} \right]_{s, s^*}$$

where $\tau = 60 \times 86400$ seconds, $s^* = 28$ is the absorbing success state, and Q^* is the generator with row s^* zeroed out (making success absorbing).

2 Model 1: Global CTMC

2.1 Overview

A single generator matrix Q is estimated from all training journeys pooled together. This is the simplest model and serves as the baseline for the others.

2.2 Fitting

Let $\mathcal{D}$ be the set of all consecutive-state transitions $(i, j, \Delta t)$ extracted from training journeys. Define:

$$N_{ij} = \#\{(i, j, \cdot) \in \mathcal{D}\}, \quad T_i = \sum_{(i, j, \Delta t) \in \mathcal{D}} \Delta t$$

The maximum-likelihood estimate of the off-diagonal rates is

$$\hat{\lambda}_{ij} = \frac{N_{ij}}{T_i}, \quad i \neq j$$

with diagonal entries set to $q_{ii} = -\sum_{j \neq i} \hat{\lambda}_{ij}$. Self-transitions are discarded before counting.

2.3 Prediction

Absorption probability. The success state $s^* = 28$ is made absorbing by zeroing row s^* in Q . For a journey currently in state s , the predicted probability is:

$$\hat{p}(s) = \left[e^{Q^* \tau} \right]_{s, s^*}$$

computed via the matrix exponential (`scipy.linalg.expm`).

Expected hitting time. For transient states $\mathcal{T} = \mathcal{S} \setminus \{s^*\}$, the expected time to absorption satisfies the linear system

$$Q_{\mathcal{T}} m = -\mathbf{1}$$

where $Q_{\mathcal{T}}$ is the sub-matrix of Q restricted to transient rows and columns. Solving gives the expected seconds to success from each state.

Fallback. If the current state of a test journey was never seen during training, the model falls back to the empirical success prior implied by the transition count table.

3 Model 2: Laplace-Smoothed Global CTMC

3.1 Overview

The global CTMC can produce noisy rates for rarely-observed state pairs because N_{ij} is often zero. `LaplaceGlobalCTMC` adds pseudo-counts to regularize the generator.

3.2 Fitting

Before computing rates, every off-diagonal count cell receives an α -smoothing prior:

$$\tilde{N}_{ij} = N_{ij} + \alpha, \quad i \neq j$$

with $\alpha = 0$ recovering the unsmoothed model, $\alpha = 1$ giving standard Laplace smoothing, and $\alpha = 0.5$ (default) corresponding to the Jeffreys prior. Rates are then computed as before:

$$\hat{\lambda}_{ij} = \frac{\tilde{N}_{ij}}{T_i}$$

3.3 Prediction

Identical to Global CTMC. Because the generator is denser (no true zeros for unseen pairs), the matrix exponential tends to assign nonzero probability to every reachable state, reducing the variance of absorption probability estimates for rare states.

4 Model 3: Clustered CTMC

4.1 Overview

Rather than pooling all journeys into one Q , customers are grouped into K behavioral clusters and a separate `GlobalCTMC` is fit per cluster. The intuition is that different customer types (e.g., browsers vs. decisive buyers) have different transition dynamics, and a per-segment Q captures this heterogeneity.

4.2 Clustering

Feature extraction. For each journey prefix, the following features are computed without using the outcome label (no leakage):

- Proportion of each of the top-20 most frequent actions (`action_prop_{ed_id}`).
- Entry state (`first_state`) and current state (`current_state`).

K-Means. Features are imputed (median) and standardized, then K -Means with $K = 3$ clusters (default) is fit on the combined feature matrix.

4.3 Fitting

Transitions are partitioned by cluster assignment, and a `GlobalCTMC` is fit independently on each partition:

$$Q^{(k)} = \text{GlobalCTMC.fit}(\mathcal{D}_k), \quad k = 1, \dots, K$$

4.4 Prediction

At test time:

1. Compute the same feature vector for the test journey.
2. Assign it to cluster k^* via the fitted K-Means pipeline.
3. Use $Q^{(k^*)}$ to compute the absorption probability.
4. If k^* has no fitted model (unseen cluster), fall back to the global model.

Blended submission. In the final Kaggle submission, the global and clustered predictions are averaged:

$$\hat{p}_{\text{blend}}(s) = 0.5 \hat{p}_{\text{global}}(s) + 0.5 \hat{p}_{\text{clustered}}(s)$$

5 Model 4: Calibrated CTMC

5.1 Overview

All CTMC models output absorption probabilities that reflect dynamics in the *training* data. Because the Kaggle test set has a much lower success prevalence ($\approx 5\%$) than training ($\approx 22\%$), raw probabilities are systematically over-confident.

`CalibratedCTMC` is a post-hoc wrapper that corrects this in two complementary ways.

5.2 Isotonic Calibration

An isotonic regression is fit on a held-out calibration fold:

$$\hat{f} = \arg \min_{f \text{ non-decreasing}} \sum_i (y_i - f(\hat{p}_i))^2$$

Because isotonic regression minimizes squared error, fitting it directly minimizes the Brier score on the calibration set — the Kaggle evaluation metric.

At test time, raw predictions are passed through $\hat{f}$:

$$\hat{p}_{\text{cal}}(s) = \hat{f}(\hat{p}_{\text{raw}}(s))$$

5.3 Prevalence Shift (Log-Odds Adjustment)

A simpler but complementary correction is applied uniformly to all model outputs before submission. Let π_{train} and π_{test} be the success rates in training and test respectively. Using a Bayes prior-probability adjustment in log-odds space:

$$r = \frac{\pi_{\text{test}}/(1 - \pi_{\text{test}})}{\pi_{\text{train}}/(1 - \pi_{\text{train}})} \approx 0.188$$

$$\hat{p}_{\text{adj}} = \frac{\hat{p} \cdot r}{\hat{p} \cdot r + (1 - \hat{p})}$$

This is applied in `calibrate_prevalence()` in `tabular_submission.py` to every model's output ($\pi_{\text{train}} = 0.2194$, $\pi_{\text{test}} = 0.05018$, estimated from the all-zeros Kaggle Brier score).

6 Model Hierarchy and Composition

The four models build on one another as follows:

Model	Builds on	Key addition
GlobalCTMC	—	Single pooled Q ; MLE rates
LaplaceGlobalCTMC	GlobalCTMC	Add- α smoothing on N_{ij}
ClusteredCTMC	GlobalCTMC	Per-segment $Q^{(k)}$ via K-Means
CalibratedCTMC	Any CTMC model	Isotonic + prevalence shift

The submission pipeline (`ctmc_submission.py`) always applies `calibrate_prevalence` to all outputs. `CalibratedCTMC` (isotonic) is used for offline Brier diagnostics in `ctmc_advanced.py`.

7 Inference Summary

All models share the same two prediction modes:

Absorption probability $\hat{p}(s) = [e^{Q^*\tau}]_{s,s^*}$ — the direct estimate of $P(\text{success within } \tau \mid \text{current state} = s)$. Used as the primary Kaggle submission score.

Expected hitting time $\hat{m}(s)$ from $Q_{\mathcal{T}m} = -\mathbf{1}$ — used experimentally in `ctmc_waiting_time_experiment.py` to compare against the absorption probability as a ranking signal (transformed as $e^{-\hat{m}/\tau}$).